DESCENT DATA VERSUS SHEAVES UNDER AN FPPF COVERING OF ALGEBRAIC SPACES: FOR AN FPPF COVERING {X_I -> X} OF ALGEBRAIC SPACES OVER S, DESCENT DATA RELATIVE TO THE COVERING ARE EQUIVALENT TO SHEAVES F ON (SCH/S)_FPPF WITH A MAP F -> X SUCH THAT EVERY X_I X_X F IS AN ALGEBRAIC SPACE; F IS AN ALGEBRAIC SPACE IFF THE CORRESPONDING DATUM IS EFFECTIVE.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3803-3835 of sources/stacks-spaces-descent/pinned-original-source.tex; proof span is lines 3837-3881, the complete author proof (43 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-spaces-descent, tightened-ruling accept list (construction + required verification in span; only the mutual-inverse check left to the reader).

Editorial note. Span computed from the \label marker to \end{lemma}; precard_gate combined span RESULT=OK (probe L9999).

Editorial note. rqual sv-mismatch fix 2026-10-14: source_version aligned to SOURCE.json (semicolon+git form, intake authority; package evidence.json already correct); revision bumped for re-assembly per publication rotation (takeover-line defer ticket, relayed by the s15 line).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 0ADV: sources/stacks-spaces-descent/pinned-original-source.tex, source lines 3881–3881. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03Y8: sources/stacks-spaces-descent/pinned-original-source.tex, source lines 3881–3881. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $\{X_i \to X\}_{i\in I}$ be an fppf covering of algebraic spaces over S (Topologies on Spaces, Definition 03Y8). There is an equivalence of categories*

$$\left\{\begin{array}{c}\text{descent data } (V_i, \varphi_{ij}) \\ \text{relative to } \{X_i \to X\}\end{array}\right\} \leftrightarrow \left\{\begin{array}{c}\text{sheaves } F \text{ on } (Sch/S)_{fppf} \text{ endowed} \\ \text{with a map } F \to X \text{ such that each} \\ X_i \times_X F \text{ is an algebraic space}\end{array}\right\}.$$

Moreover,

(1) *the algebraic space $X_i \times_X F$ on the right hand side corresponds to V_i on the left hand side, and*
(2) *the sheaf F is an algebraic space*[1] *if and only if the corresponding descent datum (X_i, φ_{ij}) is effective.*

Proof. Let us construct the functor from right to left. Let $F \to X$ be a map of sheaves on $(Sch/S)_{fppf}$ such that each $V_i = X_i \times_X F$ is an algebraic space. We have the projection $V_i \to X_i$. Then both $V_i \times_X X_j$ and $X_i \times_X V_j$ represent the sheaf $X_i \times_X F \times_X X_j$ and hence we obtain an isomorphism

$$\varphi_{ii'} : V_i \times_X X_j \to X_i \times_X V_j$$

It is straightforward to see that the maps φ_{ij} are morphisms over $X_i \times_X X_j$ and satisfy the cocycle condition. The functor from right to left is given by this construction $F \mapsto (V_i, \varphi_{ij})$.

Let us construct a functor from left to right. The isomorphisms φ_{ij} give isomorphisms

$$\varphi_{ij} : V_i \times_X X_j \longrightarrow X_i \times_X V_j$$

over $X_i \times X_j$. Set F equal to the coequalizer in the following diagram

$$\coprod_{i,i'} V_i \times_X X_j \xrightarrow[\mathrm{pr}_1 \circ \varphi_{ij}]{\mathrm{pr}_0} \coprod_i V_i \longrightarrow F$$

The cocycle condition guarantees that F comes with a map $F \to X$ and that $X_i \times_X F$ is isomorphic to V_i. The functor from left to right is given by this construction $(V_i, \varphi_{ij}) \mapsto F$.

We omit the verification that these constructions are mutually quasi-inverse functors. The final statements (1) and (2) follow from the constructions. □

[1] We will see later that this is always the case if I is not too large, see Bootstrap, Lemma 0ADV.